



Deep-sea submersibles

Scuba divers cannot dive very deep below the surface, because a diver's unprotected body cannot cope with the effects of the intense pressure in deep water. Exploring the deep ocean requires special craft called deep-sea submersibles. These are designed for scientific work on the ocean floor, and they can dive much deeper than military submarines. Some carry people, but others are remotely controlled from ships on the surface using video technology.



INTO THE DEEP

The first submersible capable of resisting the intense pressure in the ocean depths was called the bathysphere, designed in 1928 by American engineer Otis Barton. It was made of steel with 30-in- (76-cm-) thick windows, and suspended from a ship by a steel cable. Naturalist William Beebe (left) and Barton (right) used it to make the first studies of life in the twilight zone.

FULL CONTROL

Modern deep-sea exploration began with the fully controllable *Alvin*. Owned by the US Navy, but operated by the Woods Hole Oceanographic Institution, *Alvin* made its first deep dive in 1965. Like the bathysphere, *Alvin* has a spherical pressure-proof cabin, but this is mounted inside a motorized hull equipped with lights, cameras, grabs, and sample baskets. *Alvin* is still in use, and has made more than 4,600 dives—including the first manned survey of the shipwrecked ocean liner *Titanic*.

